

SANJAY B

☎ +91-8667660435 ✉ sanjayboopathy32@gmail.com 🌐 Sanjay B 📧 sanjay2902

Education

- **Madras Institute of Technology, Anna University** **October 2022 – Present**
B.Tech – Information Technology CGPA:8.2 (relative grading) Chennai, Tamil Nadu
- **The Vikasa School**
Class 12 ISC Percentage:93.75
- **The Vikasa School**
Class 10 ICSE Percentage:90

Technical Skills

- **Languages:** Python, Java, SQL
- **Web Technologies:** HTML, NodeJS, MongoDB, ReactJS
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management System, Object Oriented Methodologies, Machine Learning
- **Utilities & Framework:** GitHub, ExpressJS, Fast API, AWS, Postman

Projects

Medical Reasoning Using VLM Model for Histopathology Interpretation **December 2025**

- Built a zero-shot medical image classification system that removes dependency on annotated datasets, enabling scalable and effective cancer diagnosis support.
- Introduced an ontology-driven semantic understanding approach that translates medical knowledge into model-friendly prompts, significantly improving the model's ability achieving **92%** accuracy.
- **Technologies:** *React, Node.js, Python, VLM(Biomed CLIP)*

Document Summarization Tool **March 2025**

- Engineered an NLP pipeline using spaCy and NLTK and implemented the graph-based TextRank algorithm with NetworkX to rank and extract key sentences.
- Integrated Hugging Face Transformer model Pegasus for abstractive summarization within a robust application with comprehensive error handling.
- **Technologies:** *Python, NLTK, Transformer, NetworkX*

Connect+ **October 2024**

- Architected a full-stack community forum facilitating user engagement through text-based posts, private messaging, and voice-based interactions.
- Achieved real-time communication between concurrent users by integrating WebSockets for live messaging features.
- **Technologies:** *React, Node.js, Express, MongoDB*

Achievements

- Ranked in the **Top 3%** out of **100K+ participants** in Amazon National Coding Contest (by Scaler), earning a selection to Amazon ML Summer School 2025.
- Finalist – National Level Coding Contest AlgoChamp by **Apti-Tech** 🏆
- Runner-up – Code Wars 2024, organized by ITA, Mutex 2024
- Solved **500+** coding problems across **GeeksforGeeks** 🏆 and **LeetCode**. 🏆 Leetcode Knight rating 1875
- Represented the Tamil Nadu & Puducherry region in Kabaddi at the state level.

Position of Responsibility

Member of Event Management Team of TEDc MIT
Placement Coordinator for Department of Information Technology for the year 2025-2026